Lecture 11

Poisson manifolds and the rigid body

Objectives. *Lecture 10 divided symmetries out of canonical systems. Dividing the rotations out of the rigid body completely leaves three variables, the components of its angular momentum, with a bracket whose matrix $\Pi(x)$ varies from point to point and is singular everywhere. What survives when the bracket is such a matrix? A union of phase spaces: for the angular-momentum bracket, the spheres centred at the origin, and the origin. The body angular momentum $\mathbf{M}$ of the free rigid body moves on those spheres, and on each of them its rotations about the axes of least and greatest inertia are stable and the rotation about the middle axis is not.*

11.1 Poisson structures on $\mathbb{R}^m$

Since Lecture 5 a Hamiltonian system has lived on an open subset of $\mathbb{R}^{2n}$ with canonical coordinates, $\{q_i, p_j\} = \delta_{ij}$ and all other brackets zero. We now keep the algebra of Lecture 5 — bilinear, antisymmetric, Leibniz, Jacobi — and drop the coordinates.

Definition 11.1 (Poisson structure)**.** Let $U \subseteq \mathbb{R}^m$ be open. A *Poisson structure* on U is a smooth map $\Pi\colon U \to \mathbb{R}^{m\times m}$ with $\Pi(x)^{\mathsf{T}} = -\Pi(x)$ for every x, such that the bracket

$$\{f, g\}(x) = \nabla f(x)^{\mathsf{T}}\,\Pi(x)\,\nabla g(x) = \sum_{a,b=1}^{m} \Pi_{ab}(x)\,\frac{\partial f}{\partial x_a}\frac{\partial g}{\partial x_b} \tag{11.1}$$

satisfies the Jacobi identity on $C^\infty(U)$. We call Π the *structure matrix* and (U, Π) a *Poisson manifold*. The *Hamiltonian field* of f is $X_f = \Pi\nabla f$.

Three things are new. The dimension m need not be even. The matrix $\Pi(x)$ need not be invertible. And Π may vary from point to point. The old bracket is the case $m = 2n$, $\Pi = \mathbb{J}$.

Everything formal is unchanged. Putting $f = x_a$, $g = x_b$ in (11.1) gives

$$\{x_a, x_b\} = \Pi_{ab}, \tag{11.2}$$

so the structure matrix *is* the table of brackets of the coordinates. Bilinearity and antisymmetry are immediate, and so is Leibniz ($\{f, gh\} = \{f, g\}\,h + g\,\{f, h\}$), because (11.1) differentiates g once. Only the Jacobi identity is a condition, and by Remark 5.6 it holds for all observables as soon as it holds for the coordinate functions. As in Definition 6.1, $X_f h = \nabla h^{\mathsf{T}}\Pi\nabla f = \{h, f\}$, so the motion generated by H is

$$\dot{x} = \Pi(x)\,\nabla H(x), \qquad \dot{f} = \{f, H\}, \tag{11.3}$$

and H is conserved along it, since $\{H, H\} = 0$ by antisymmetry.

An invertible $\Pi(x)$ forces m even, since $\det\Pi = \det\Pi^{\mathsf{T}} = (-1)^m \det\Pi$. This lecture is about the other case, a *singular* $\Pi(x)$.

11.2 The Lie–Poisson bracket, and where it comes from

Here is the bracket this lecture is about.

Definition 11.2 (The Lie–Poisson bracket)**.** On $\mathbb{R}^3$, with the point written $\mathbf{L} = (L_1, L_2, L_3)$, the *Lie–Poisson bracket* is

$$\{f, g\}(\mathbf{L}) = \mathbf{L} \cdot (\nabla f \times \nabla g), \qquad \text{that is} \qquad \Pi_{ab}(\mathbf{L}) = \sum_c \varepsilon_{abc} L_c. \tag{11.4}$$

The structure matrix is *linear* in $\mathbf{L}$, and its coordinate brackets are

$$\{L_1, L_2\} = L_3, \qquad \{L_2, L_3\} = L_1, \qquad \{L_3, L_1\} = L_2, \tag{11.5}$$

which is (5.11): the brackets of the three components of angular momentum. From (11.4), $X_f = \Pi \nabla f$ works out to

$$X_f(\mathbf{L}) = \nabla f(\mathbf{L}) \times \mathbf{L}. \tag{11.6}$$

In particular $X_{L_3} = e_3 \times \mathbf{L}$: *the third component generates the rotation about the third axis*, counter-clockwise seen from above and at unit angular speed.

Proposition 11.3 (The Lie–Poisson bracket is a Poisson structure)**.** (11.4) *satisfies the Jacobi identity.*

Proof. By Remark 5.6 it is enough to check it on the coordinates. For constant $u, v \in \mathbb{R}^3$, (11.4) gives $\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\} = (u \times v) \cdot \mathbf{L}$, so the cyclic sum of $\{\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\}, w \cdot \mathbf{L}\}$ is $\big[(u \times v) \times w + (v \times w) \times u + (w \times u) \times v\big] \cdot \mathbf{L} = 0$, by the Jacobi identity of the cross product. Take $u, v, w = e_a, e_b, e_c$. ∎

This bracket is what the canonical bracket leaves behind when only the angular momentum is kept. To say so we need maps that carry one bracket to another.

Definition 11.4 (Poisson map)**.** Let (U, Π) and (V, Π') be Poisson manifolds. A smooth $F : U \to V$ is a *Poisson map* if $\{f \circ F, g \circ F\}_U = \{f, g\}_V \circ F$ for all $f, g \in C^\infty(V)$.

Theorem 11.5 ($q \times p$ is a Poisson map)**.** *$J : \mathbb{R}^6 \to \mathbb{R}^3$, $J(q, p) = q \times p$, is a Poisson map from the canonical bracket to* (11.4).

Proof. By the chain rule, for $f \in C^\infty(\mathbb{R}^3)$, $\partial(f \circ J)/\partial q_i = \sum_a (\partial_a f \circ J)\, \partial J_a / \partial q_i$, and likewise in p. Substituting into (5.1) and collecting,

$$\{f \circ J, g \circ J\} = \sum_{a,b} (\partial_a f \circ J)(\partial_b g \circ J)\ \{J_a, J_b\}.$$

By (5.11), $\{J_a, J_b\} = \sum_c \varepsilon_{abc} J_c$, so the right-hand side is $\big[\mathbf{L} \cdot (\nabla f \times \nabla g)\big] \circ J$ — the function of $\mathbf{L}$ in the square bracket, evaluated at $\mathbf{L} = J(q, p)$ — which is $\{f, g\} \circ J$ for the bracket (11.4). ∎

11.3 Casimirs and leaves

Definition 11.6 (Casimir). $C \in C^\infty(U)$ is a *Casimir* of a Poisson structure if $\{C, f\} = 0$ for every $f \in C^\infty(U)$, equivalently if $X_C = 0$.

A Casimir is conserved by *every* Hamiltonian, since $\dot C = \{C, H\} = 0$ whatever H is, and it generates nothing: its flow is the identity. It is the opposite of a symmetry charge, which is conserved by one H and generates a motion. On a phase space there are no interesting Casimirs: Proposition 5.8 showed that on a connected canonical phase space $\{C, f\} = 0$ for all f forces C constant, and the same argument works wherever Π is invertible, since $X_C = \Pi\nabla C = 0$ then gives $\nabla C = 0$. Degeneracy is what makes the notion non-trivial.

Proposition 11.7 (The Casimirs of (11.4)). *$C \in C^\infty(\mathbb{R}^3)$ is a Casimir of the Lie–Poisson bracket if and only if $\mathbf{L} \times \nabla C(\mathbf{L}) = 0$ for every $\mathbf{L}$, and that holds if and only if C is constant on every sphere $|\mathbf{L}| = R$. In particular $|\mathbf{L}|^2$ is a Casimir.*

Proof. By the triple-product identity, $\{C, f\} = \mathbf{L} \cdot (\nabla C \times \nabla f) = \nabla f \cdot (\mathbf{L} \times \nabla C)$. This vanishes for every f if and only if $\mathbf{L} \times \nabla C = 0$ (take $f = L_a$ for the three values of a). Suppose that holds. At $\mathbf{L} \neq 0$ it says $\nabla C(\mathbf{L})$ is a multiple of $\mathbf{L}$, so the derivative of C in every direction tangent to the sphere through $\mathbf{L}$ is zero; joining two points of that sphere by a smooth path on it and differentiating C along the path shows C is constant on it. The sphere of radius 0 is a point. Conversely, if C is constant on every sphere then its derivative along every tangent direction vanishes, so ∇C is radial and $\mathbf{L} \times \nabla C = 0$. Finally $\nabla |\mathbf{L}|^2 = 2\mathbf{L}$. ∎

So every motion, whatever its Hamiltonian, stays on the sphere $|\mathbf{L}| = R$ on which it starts. That sphere ought to be a phase space with one degree of freedom. But it is not an open subset of $\mathbb{R}^2$, and no single system of coordinates covers it.

Phase space by canonical charts

By a *chart* on a set $S \subseteq \mathbb{R}^N$ we mean a bijection $\kappa : U \to V$ from a subset $U \subseteq S$ that is *open in S* (of the form $S \cap O$ with $O \subseteq \mathbb{R}^N$ open) onto an open set $V \subseteq \mathbb{R}^k$, continuous in both directions.

Proposition 11.8 (No chart covers the sphere). *Let $S^2_R = \{\mathbf{L} \in \mathbb{R}^3 : |\mathbf{L}| = R\}$ with $R > 0$. There is no chart $\kappa : S^2_R \to V$ with $V \subseteq \mathbb{R}^2$ open.*

Proof. The first component κ_1 of κ is a continuous function on the closed, bounded set $S^2_R \subseteq \mathbb{R}^3$, so it attains a maximum at some point P. But V is open, so it contains $\kappa(P) + (\varepsilon, 0)$ for small $\varepsilon > 0$, and this is κ of a point where κ_1 is larger. ∎

The remedy is that of Lecture 2: overlapping patches that agree where they overlap. One level up, "agree" means *canonically*.

Definition 11.9 (Phase space). Let $\mathcal{M} \subseteq \mathbb{R}^N$. A *canonical atlas* of dimension $2n$ on $\mathcal{M}$ is a family of charts $\kappa_a : U_a \to V_a \subseteq \mathbb{R}^{2n}$ whose domains cover $\mathcal{M}$, such that for every pair a, b the *transition map* $\tau_{ba} = \kappa_b \circ \kappa_a^{-1} : \kappa_a(U_a \cap U_b) \to \kappa_b(U_a \cap U_b)$ is smooth and canonical (Definition 8.1). The coordinates (q, p) of a chart are *canonical coordinates*. A set with a canonical atlas is a *phase space*, or *symplectic manifold*.

Now the sphere. Away from the poles $(0,0,\pm R)$ use Archimedes' coordinates, the *azimuth* φ and the *height* $z = L_3$:

$$\mathbf{L} = \big(\rho\cos\varphi,\ \rho\sin\varphi,\ z\big), \qquad \rho = \sqrt{R^2 - z^2}, \qquad |z| < R. \tag{11.7}$$

An angle needs two ranges, $(-\pi,\pi)$ and $(0,2\pi)$, to reach every azimuth, so "the chart (φ, z)" is really two charts differing by the shift $\varphi \mapsto \varphi + 2\pi$; we shall not mention this again. The second chart does the same about the first axis: the height is L_1, and ψ is the angle about the first axis measured from e_2 towards e_3, so that $\mathbf{L} = (L_1, \rho\cos\psi, \rho\sin\psi)$ with $\rho = \sqrt{R^2 - L_1^2}$. It is the first chart composed with the cyclic relabelling $(L_1, L_2, L_3) \mapsto (L_2, L_3, L_1)$, and it misses only $(\pm R, 0, 0)$, so between them these charts cover the sphere.

Theorem 11.10 (The spheres are phase spaces, and the bracket is theirs · *proved; the completeness in (1) uses T1 (cited)*)**.** *Let $R > 0$. For the Lie–Poisson bracket* (11.4)*:*

1. *every Hamiltonian flow maps S^2_R to itself and is complete there;*
2. *for $\mathbf{L} \in S^2_R$ and f, g smooth near $\mathbf{L}$, the value $\{f,g\}(\mathbf{L})$ depends only on the restrictions of f and g to the sphere through $\mathbf{L}$;*
3. *in the chart (φ, z) this bracket is the canonical bracket of $q = \varphi,\ p = z$, so that*

$$\{\varphi, z\} = +1, \tag{11.8}$$

   and in the chart (ψ, L_1) it is the canonical bracket of $q = \psi,\ p = L_1$. These charts are therefore a canonical atlas, and S^2_R is a phase space of dimension 2;
4. *$\Pi(0) = 0$: every X_f vanishes at the origin, which is therefore an orbit by itself.*

So $\mathbb{R}^3$, which is odd-dimensional and has no canonical coordinates, is the disjoint union of the spheres and the origin, and on each piece the Lie–Poisson bracket is the bracket of a phase space. These pieces are the symplectic leaves*, and here the Casimir $|\mathbf{L}|^2$ labels them.*

Proof. (1) $|\mathbf{L}|^2$ is a Casimir (Proposition 11.7), so it is conserved by every flow, and a motion that starts on S^2_R stays on it as long as it exists; S^2_R is compact, so by the blow-up alternative of Theorem 1.3 the motion exists for all time, and Theorem 1.5 makes the flow a one-parameter group of maps of S^2_R.

(2) $\mathbf{L}\cdot(\nabla f \times \nabla g)$ is unchanged if ∇f or ∇g is altered by a multiple of $\mathbf{L}$. Two smooth functions that agree on the sphere through $\mathbf{L}$ have gradients at $\mathbf{L}$ that differ by such a multiple, since their difference vanishes on the sphere.

(3) The azimuth φ and the height L_3 are also smooth functions on an open subset of $\mathbb{R}^3$, with $\nabla\varphi = (-L_2, L_1, 0)/(L_1^2 + L_2^2)$, so by (11.4)

$$\{\varphi, L_3\} = \mathbf{L}\cdot\big(\nabla\varphi \times e_3\big) = \mathbf{L}\cdot\frac{(L_1, L_2, 0)}{L_1^2 + L_2^2} = 1.$$

Let f, g have local expressions $F(\varphi, z)$, $G(\varphi, z)$ on the sphere. By (2) we may replace f by $F(\varphi, L_3)$ and g by $G(\varphi, L_3)$, and then the chain rule and antisymmetry give

$$\{f,g\} = \left(\frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z} - \frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi}\right)\{\varphi, L_3\} = \frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z} - \frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi},$$

the canonical bracket of F and G. The same computation with the indices relabelled cyclically, $1 \to 2 \to 3 \to 1$, which leaves ε_{abc} unchanged, gives $\{\psi, L_1\} = 1$ and the same conclusion in the second chart.

For the atlas, the two ranges of φ (or of ψ) differ by a translation, which is canonical. The transition τ from (φ, z) to (ψ, L_1) is smooth, and its two components are the local expressions of ψ and L_1 in the first chart. By what was just shown their canonical bracket is $\{\psi, L_1\} = 1$, which is condition (iv) of Theorem 8.2 for $n = 1$; so τ is canonical. The transition the other way is canonical by the same argument with $\{\varphi, L_3\} = 1$.

(4) is (11.4) at $\mathbf{L} = 0$. ∎

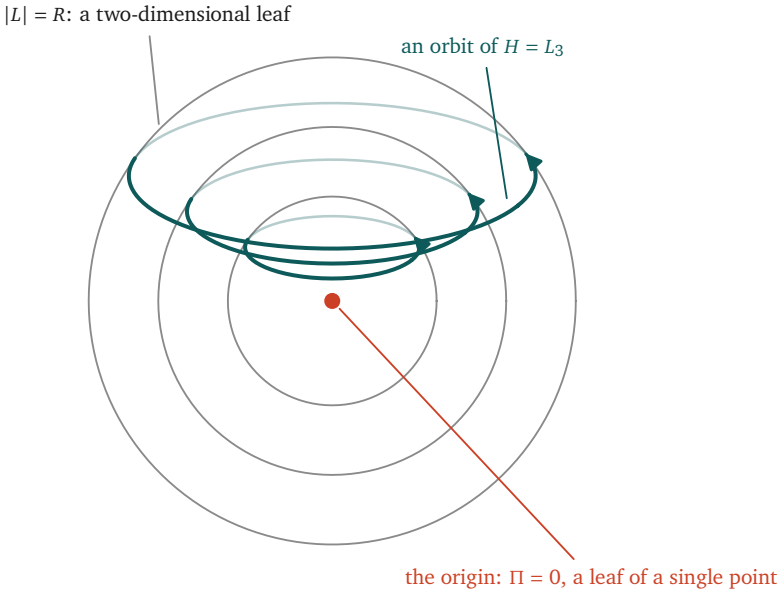


Figure 11.1: The symplectic leaves of (11.4): the spheres $|\mathbf{L}| = R$, each a two-dimensional phase space, and the origin, a leaf consisting of one point, where $\Pi = 0$. On each sphere one orbit of $H = L_3$ is drawn; it was integrated from $\dot{\mathbf{L}} = \nabla H \times \mathbf{L}$, not traced as a circle of latitude, and $|\mathbf{L}|$ and L_3 are constant along it to 10^{-10}.

11.4 The rigid body and its sign

A rigid body with a point fixed at the origin has configuration space SO(3) (§ 2.2), so its phase space $T^*\mathrm{SO}(3)$ is six-dimensional. Its kinetic energy is unchanged if the whole body is rotated about the fixed point, and for a free body the three components of its spatial angular momentum are conserved (§ 11.5 sketches why). Keeping only the angular momentum leaves three variables and a degenerate bracket on $\mathbb{R}^3$, as $q \times p$ did. We state the result and do not prove it: nothing in this course builds $T^*\mathrm{SO}(3)$.

Proposition 11.11 (Reduction of the rigid body · *cited*)**.** *Dividing the rotations out of the rigid body's phase space leaves the* body *angular momentum* $\mathbf{M}$ *as its state, with the Poisson structure* $(\mathbb{R}^3, -\Pi)$*, that is,*

$$\{f, g\}(\mathbf{M}) = -\mathbf{M} \cdot (\nabla f \times \nabla g). \tag{11.9}$$

Read in the spatial *angular momentum* $\mathbf{L}$ *instead, the bracket is* (11.4)*, as for* $q \times p$*. (Marsden–Ratiu §§13.1–13.4 and §15.3, pp. 489–491; their conventions are ours.)*

The minus sign is the trap of this lecture. The two brackets are mirror images, not rivals: $\mathbf{L} \mapsto -\mathbf{L}$ is a Poisson map from (11.4) to (11.9), as one line of (11.4) shows. And the sign is not taken on trust: § 11.5 derives the equation of motion of $\mathbf{M}$ from rigid-body kinematics, with no bracket anywhere in the argument, and only (11.9) reproduces it.

Corollary 11.12 (The body leaves)**.** *For the body bracket* (11.9)*,* $X_f = \mathbf{M} \times \nabla f$*, and the Casimirs, the leaves and their canonical charts are those of Proposition 11.7 and Theorem 11.10, carried over by* $\mathbf{L} \mapsto -\mathbf{L}$*, which maps each sphere to itself. In particular* $\{\varphi, M_3\} = -1$*: the canonical chart is* $(\varphi, -M_3)$*.*

Proof. $X_f = -\Pi \nabla f = -\nabla f \times \mathbf{M}$ by (11.6). A Poisson bijection carries Casimirs to Casimirs, flows to flows and canonical charts to canonical charts, and $-\mathbf{L}$ lies on the sphere of $\mathbf{L}$; and $\varphi(-\mathbf{L}) = \varphi(\mathbf{L}) \pm \pi$, $L_3(-\mathbf{L}) = -L_3(\mathbf{L})$. ∎

11.5 Euler's equations

The body variables

Let the body have a point fixed at the origin and be made of mass points m_a at positions X_a *in the body*, not all on one line through the origin. Its *attitude* at time t is the rotation $R(t) \in \mathrm{SO}(3)$ that carries the reference placement to the current one, so the spatial positions are $x_a(t) = R(t) X_a$. Differentiating $R^{\mathsf{T}} R = \mathbb{1}$ gives $\dot{R}^{\mathsf{T}} R + R^{\mathsf{T}} \dot{R} = 0$, so $R^{\mathsf{T}} \dot{R}$ is antisymmetric, and by Proposition 10.2 there is a unique $\Omega(t) \in \mathbb{R}^3$ — the *body angular velocity* — with

$$R^{\mathsf{T}} \dot{R} = \hat{\Omega}, \qquad \hat{\Omega} u = \Omega \times u, \qquad \text{hence} \qquad \dot{x}_a = R\,(\Omega \times X_a). \tag{11.10}$$

Lemma 11.13 (The inertia tensor)**.** $\mathbb{I}\Omega := \sum_a m_a X_a \times (\Omega \times X_a)$ *is a symmetric linear map of* $\mathbb{R}^3$*, positive definite because the* X_a *do not all lie on one line through the origin, and*

$$T = \tfrac{1}{2} \sum_a m_a |\dot{x}_a|^2 = \tfrac{1}{2}\, \Omega \cdot \mathbb{I}\Omega, \qquad \mathbf{L} := \sum_a m_a\, x_a \times \dot{x}_a = R\, \mathbb{I}\Omega.$$

Proof. By the triple-product identity $u \cdot \mathbb{I} v = \sum_a m_a (u \times X_a) \cdot (v \times X_a)$: symmetric, and positive at $u = v \neq 0$ unless every X_a lies on $\mathbb{R}u$. Then $|\dot{x}_a| = |\Omega \times X_a|$ gives T, and $Ru \times Rv = R(u \times v)$ gives $\mathbf{L}$. ∎

Write $\mathbf{M} := R^{\mathsf{T}} \mathbf{L} = \mathbb{I}\Omega$ for the *body* angular momentum. In principal axes — the orthonormal eigenbasis of $\mathbb{I}$ — $\mathbb{I} = \mathrm{diag}(I_1, I_2, I_3)$ with $I_i > 0$, so

$$M_i = I_i \Omega_i, \qquad T = \tfrac{1}{2} \sum_i I_i \Omega_i^2 = \frac{1}{2} \sum_i \frac{M_i^2}{I_i}. \tag{11.11}$$

Proposition 11.14 (Euler's equation, from kinematics)**.** *Let* $R(t) \in \mathrm{SO}(3)$ *be a smooth motion whose spatial angular momentum* $\mathbf{L} = R\, \mathbb{I}\Omega$ *is constant. Then* $\mathbf{M} = R^{\mathsf{T}} \mathbf{L}$ *satisfies*

$$\dot{\mathbf{M}} = \mathbf{M} \times \Omega, \qquad \Omega_i = M_i / I_i. \tag{11.12}$$

Proof. $\dot{\mathbf{M}} = \dot{R}^{\mathsf{T}} \mathbf{L} = (R\hat{\Omega})^{\mathsf{T}} \mathbf{L} = \hat{\Omega}^{\mathsf{T}} R^{\mathsf{T}} \mathbf{L} = -\hat{\Omega} \mathbf{M} = -\Omega \times \mathbf{M} = \mathbf{M} \times \Omega$, using (11.10) and the antisymmetry of $\hat{\Omega}$. ∎

The hypothesis is the only physical input, and we sketch why it holds for a free body. Let no external force act except the one that holds the fixed point, which has no torque about that point. The internal forces of a rigid body cancel in pairs along the lines joining the points — Newton's third law in its strong form — so the total torque about the fixed point is zero and $\mathbf{L}$ is constant.

Euler's equations are Lie–Poisson

Theorem 11.15 (Euler's equations). *On $\mathbb{R}^3$ with the body bracket (11.9) and*

$$H(\mathbf{M}) = \frac{1}{2}\Big(\frac{M_1^2}{I_1} + \frac{M_2^2}{I_2} + \frac{M_3^2}{I_3}\Big),$$

the evolution $\dot f = \{f, H\}$ is exactly (11.12), that is

$$\dot M_1 = \frac{I_2 - I_3}{I_2 I_3} M_2 M_3, \qquad \dot M_2 = \frac{I_3 - I_1}{I_3 I_1} M_3 M_1, \qquad \dot M_3 = \frac{I_1 - I_2}{I_1 I_2} M_1 M_2. \tag{11.13}$$

The spatial bracket (11.4) and the same H would give $\dot{\mathbf{M}} = \Omega \times \mathbf{M}$, the time reverse.

Proof. By Corollary 11.12 the Hamiltonian field is $\mathbf{M} \times \nabla H$, and $\nabla H = \Omega$ by (11.11). Writing out $\mathbf{M} \times \Omega$ componentwise gives (11.13). With Π in place of $-\Pi$ the field is $\nabla H \times \mathbf{M} = \Omega \times \mathbf{M}$ by (11.6). ∎

So the free rigid body is a Hamiltonian system on $\mathbb{R}^3$, with a degenerate bracket and three equations in place of six. It has two conserved quantities of different kinds: the Casimir $|\mathbf{M}|^2$, conserved because of the bracket, and H, conserved because it is the Hamiltonian. The motion runs along the intersection of a sphere with an energy ellipsoid.

Proposition 11.16 (The six equilibria). *Let $I_1 < I_2 < I_3$ and $R > 0$. The equilibria of (11.12) on the leaf $|\mathbf{M}| = R$ are exactly the six points $\pm Re_1, \pm Re_2, \pm Re_3$: the rotations about the principal axes.*

Proof. The a-th component of $\mathbf{M} \times \Omega$ is $\big(1/I_c - 1/I_b\big) M_b M_c$ with (a, b, c) cyclic. The three moments are distinct, so all three components vanish if and only if $M_b M_c = 0$ for each of the three pairs, that is, if and only if at most one M_i is non-zero. On $|\mathbf{M}| = R$ that leaves the six listed points. ∎

11.6 The stable axes

Each leaf is a two-dimensional phase space, so on a leaf the rigid body has one degree of freedom, and the question of Lecture 1 can be asked: which equilibria are stable? Throughout this section and the next $I_1 < I_2 < I_3$ and $R > 0$, and "stable on the leaf" is Definition 1.13 with neighbourhoods taken in the sphere S_R^2.

Theorem 11.17 (The axes of least and greatest inertia are stable on the leaf · *proved; the completeness uses T1 (cited)*). *On S_R^2 the function H attains its maximum $R^2/(2I_1)$ exactly at $\pm Re_1$ and its minimum $R^2/(2I_3)$ exactly at $\pm Re_3$. Each of these four points is a stable equilibrium on the leaf.*

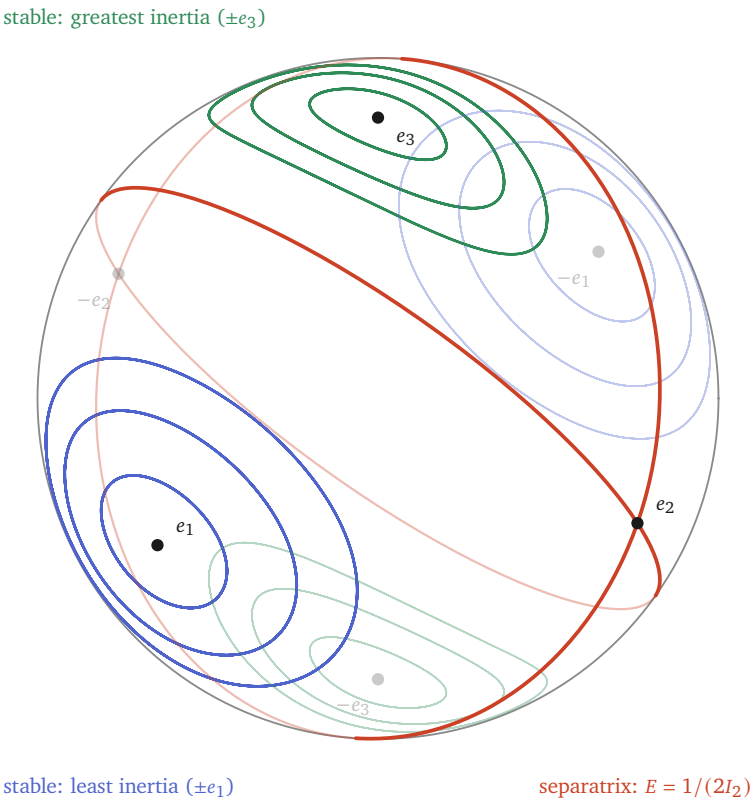


Figure 11.2: The free rigid body on the leaf $|\mathbf{M}| = 1$, with $I = (1, 2, 3)$. Every blue and green curve is an orbit, obtained by integrating (11.12) and not by intersecting the two quadrics; $|\mathbf{M}|$ and H are constant along each to 10^{-8}. Blue: the orbits round $\pm e_1$, the axis of *least* inertia, where H is largest. Green: the orbits round $\pm e_3$, of *greatest* inertia, where H is smallest. Red: the level $H = 1/(2I_2)$, two great circles drawn from their own equation $M_3 = \pm kM_1$ (§ 11.7), both passing through the two equilibria $\pm e_2$. The far hemisphere is faint.

Proof. On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_1} = \sum_i M_i^2\Big(\frac{1}{I_i} - \frac{1}{I_1}\Big) = M_2^2\Big(\frac{1}{I_2} - \frac{1}{I_1}\Big) + M_3^2\Big(\frac{1}{I_3} - \frac{1}{I_1}\Big) \le 0,$$

with equality only if $M_2 = M_3 = 0$, that is at $\pm Re_1$; and symmetrically $2H - R^2/I_3 = M_1^2(1/I_1 - 1/I_3) + M_2^2(1/I_2 - 1/I_3) \ge 0$ with equality only at $\pm Re_3$. The four points are equilibria by Proposition 11.16.

Stability is Dirichlet's argument: a conserved function with a strict extremum traps the motion. Take $x_* = Re_1$ and $\varepsilon \in (0, 2R)$; a smaller δ serves a larger ε, so this is no restriction. The set $K = \{\mathbf{M} \in S_R^2 : |\mathbf{M} - x_*| = \varepsilon\}$ is closed and bounded and contains neither Re_1 nor $-Re_1$, so $H < R^2/(2I_1)$ on K and its maximum m there is still less than $R^2/(2I_1)$. By continuity choose $\delta \in (0, \varepsilon)$ with $H > m$ at every point of the sphere within δ of x_*. A motion starting at such a point stays on the sphere and exists for all time (Theorem 11.10 (1), with Corollary 11.12), and H is conserved along it, so $H > m$ for all t and the motion never meets K. Its distance from x_* is continuous in t and starts below ε, so it stays below ε for all $t \ge 0$. The same argument works at $-Re_1$, and at $\pm Re_3$ with the inequalities reversed. ∎

11.7 The unstable axis

At the middle axis H has neither a maximum nor a minimum on the sphere. On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_2} = M_1^2\Big(\frac{1}{I_1} - \frac{1}{I_2}\Big) - M_3^2\Big(\frac{1}{I_2} - \frac{1}{I_3}\Big),$$

which takes both signs arbitrarily near $\pm Re_2$ and vanishes exactly on the two great circles $\Gamma_\pm = \{|\mathbf{M}| = R,\ M_3 = \pm k M_1\}$, with $k = \sqrt{(1/I_1 - 1/I_2)/(1/I_2 - 1/I_3)} > 0$, both through $\pm Re_2$ (Figure 11.2).

Corollary 11.18 (The middle axis is unstable · *sketched*)**.** *Neither $+Re_2$ nor $-Re_2$ is a stable equilibrium on the leaf.*

Sketch. On Γ_-, (11.13) gives $\dot{M}_2 = (1/I_1 - 1/I_3) M_3 M_1 = -(1/I_1 - 1/I_3) k\, M_1^2$, which is negative except where $M_1 = 0$, that is, at $\pm Re_2$. So a motion that starts on Γ_- near Re_2 — where it stays, since it keeps the value of H and can pass to Γ_+ only through an equilibrium — moves away from Re_2 with M_2 decreasing. That it then tends to $-Re_2$ is the step not proved here (M_2 has a limit, which can only be at an equilibrium); granting it, starting points arbitrarily close to Re_2 lead arbitrarily close to $-Re_2$, at distance $2R$. On Γ_+ the same holds with the signs reversed, at $-Re_2$. ∎

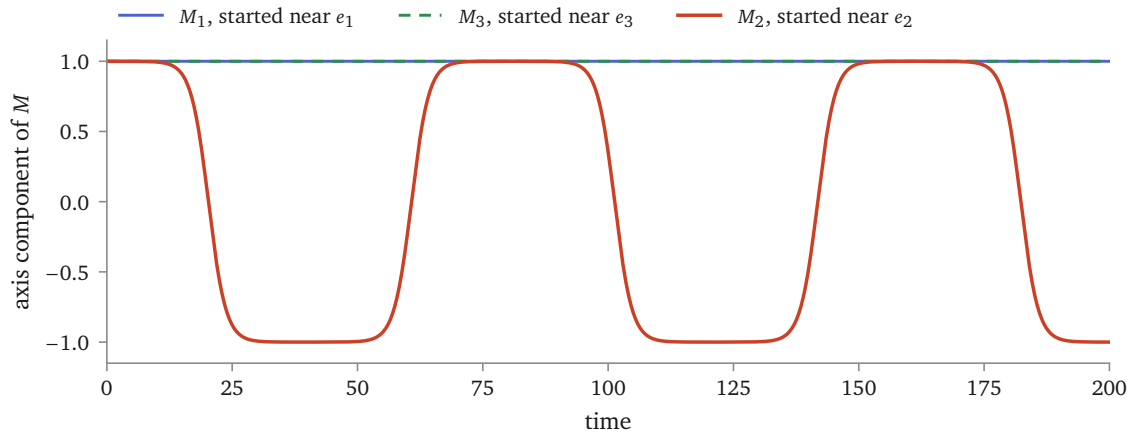


Figure 11.3: $I = (1, 2, 3)$, $|\mathbf{M}| = 1$, each run started close to a principal axis. The axis component of $\mathbf{M}$ stays close to 1 for the two extreme axes (the e_3 run dashed, since the two curves nearly coincide); about the middle axis it changes sign five times in 200 time units.

Near Re_2 but off $\Gamma_\pm$ the motion runs close beside them, from near one middle-axis equilibrium to near the other and back (Figure 11.3). In space $\mathbf{L}$ stays fixed and the body turns over, again and again: the *tennis-racket* effect. A racket or a book thrown spinning nearly about its middle axis turns over in the air if the flight lasts long enough; about the other two axes it does not.

Summary.

- *Handed on:* Poisson structures (Definition 11.1); the Lie–Poisson bracket and its Casimirs (Proposition 11.7); its leaves, each sphere a phase space covered by canonical charts and by no single chart (Definition 11.9, Theorem 11.10, and Proposition 11.8); the free rigid

body, $\dot{\mathbf{M}} = \mathbf{M} \times \Omega$ for the body bracket (Theorem 11.15), on each sphere stable about its extreme axes and not about the middle one (Theorem 11.17 and Corollary 11.18).

- *Next:* Lecture 12 asks when enough conserved quantities make a motion integrable, and gives the last answer to the course's question.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, whose conventions are ours: §1.2 (pp. 7–9) for the rigid-body bracket and Euler's equations; Chapter 10 (p. 327 on) for Poisson manifolds, with Casimir functions on p. 335; §15.3 (pp. 489–491) for the free rigid body and the flow on the angular-momentum spheres (Figure 15.3.1). Arnold, *Mathematical Methods of Classical Mechanics*, Appendix 14, p. 456: here "the leaves are concentric spheres and their center at the origin". Tong, *Lectures on Classical Dynamics*, §3.4.3 (pp. 57–58), for the tennis racket.